

TWO PORT NETWORK

→ A pair of terminals at which signal may enter or leave from the N/w is called as port.



Port

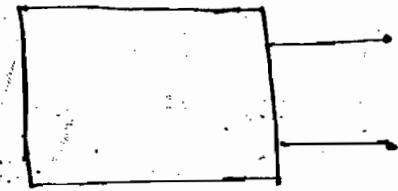
A pair of terminals

Single Port

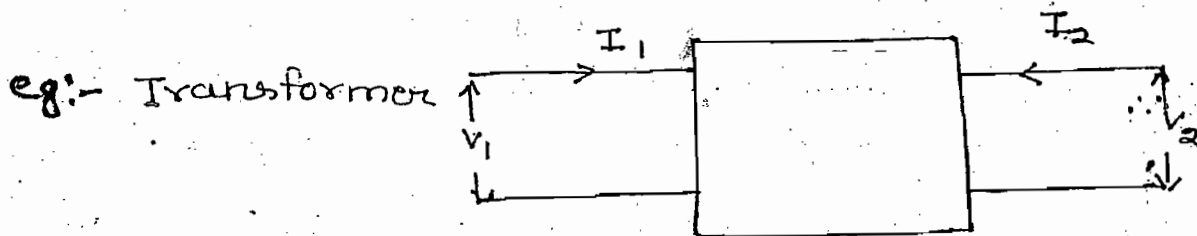
eg:- Motor

→ When the N/w is having one pair of terminals then the N/w is called as single port N/w

eg:- Motor, Generator



eg:- Generator



eg:- Transformer

Classification of Parameters:-

- (i) z - Parameter (O.C Parameter)
- (ii) y - Parameter (S.C Parameter)
- (iii) h - Parameter (Hybrid Parameter)
- (iv) g - Parameter (Inverse hybrid Parameter)
- (v) ABCD - Parameter (Transmission line parameter)
- (vi) abcd - Parameter (Inverse transmission line parameter)

(i) Z-Parameter :-

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{---(I)} \rightarrow \text{KVL}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{---(II)} \rightarrow \text{KVL}$$

$V_1, V_2 \rightarrow$ Dependent Variables

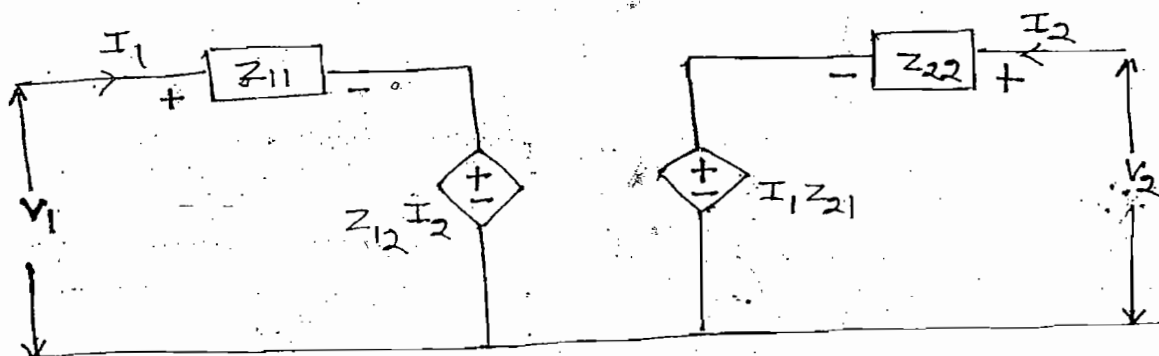
Source $\rightarrow I_1, I_2 \rightarrow$ Independent Variables

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0}$$

$$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$$

$$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$$

$$Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$



Z_{11} = Open ckt i/p impedance / Driving point i/p impedance

Z_{21} = Forward transfer impedance

Z_{12} = Reverse transfer impedance

Z_{22} = Open ckt o/p impedance / Driving point o/p impedance

(ii) Y-Parameter :-

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \text{---(I)} \rightarrow \text{KCL}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \text{---(II)} \rightarrow \text{KCL}$$

$I_1, I_2 \rightarrow$ Dependent Variables

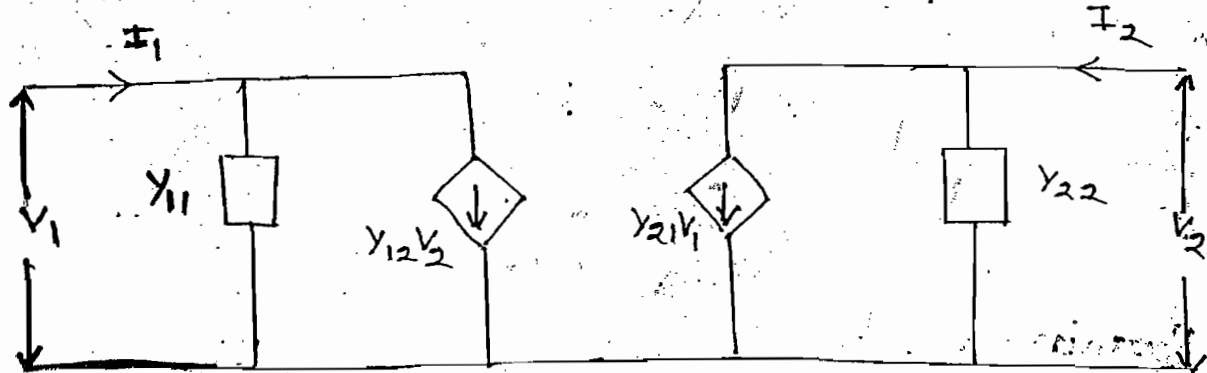
$V_1, V_2 \rightarrow$ Independent Variables
(source)

$$Y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0}$$

$$Y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0}$$

$$Y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0}$$

$$Y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0}$$



$Y_{11} =$ S.C i/p admittance / Driving point i/p admittance

$Y_{21} =$ forward transfer admittance

$Y_{12} =$ Reverse transfer admittance

$Y_{22} =$ S.C o/p admittance / Driving point o/p admittance

h-Parameter:-

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \text{--- (I) } \rightarrow \text{KVL}$$

$$I_2 = h_{21}I_1 + h_{22}V_2 \quad \text{--- (II) } \rightarrow \text{KCL}$$

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0} \quad \left(h_{11} \neq Z_{11} \quad , \quad h_{11} = \frac{1}{Y_{11}} \right)$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0} \quad \left(h_{21} = \frac{Y_{21}}{Y_{11}} = \frac{I_2/V_1}{I_1/V_1} \right)$$

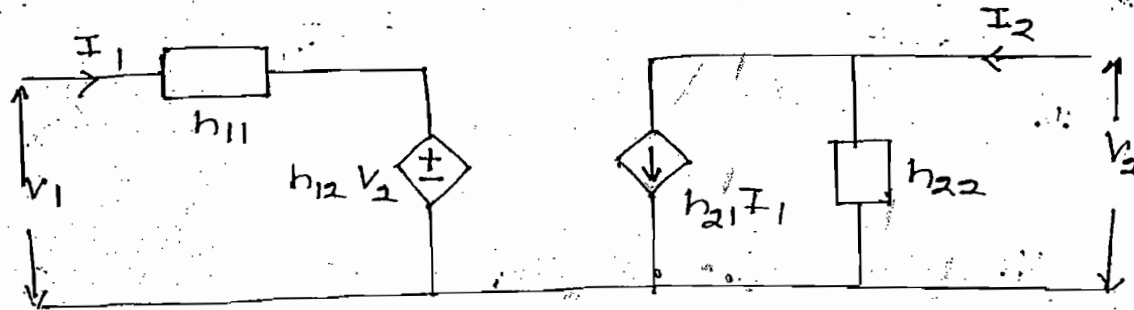
(unitless)

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0} \quad \left(h_{12} = \frac{Z_{12}}{Z_{22}} \right)$$

↓
Unitless

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0} \quad \left(h_{22} \neq Y_{22}, \quad h_{22} = \frac{1}{Z_{22}} \right)$$

↓
(mho)



g-Parameters:-

$$I_1 = g_{11}V_1 + g_{12}I_2 \quad \text{--- (I)}$$

$$V_2 = g_{21}V_1 + g_{22}I_2 \quad \text{--- (II)}$$

$$g_{11} = \left. \frac{I_1}{V_1} \right|_{I_2=0}$$

↓
mho

$$\left(g_{11} \neq Y_{11}, \quad g_{11} = \frac{1}{Z_{11}} \right)$$

$$g_{21} = \left. \frac{V_2}{V_1} \right|_{I_2=0}$$

↓
Unitless

$$\left(g_{21} = \frac{Z_{21}}{Z_{11}} \right)$$

$$g_{12} = \left. \frac{I_1}{I_2} \right|_{V_1=0}$$

↓
(Unitless)

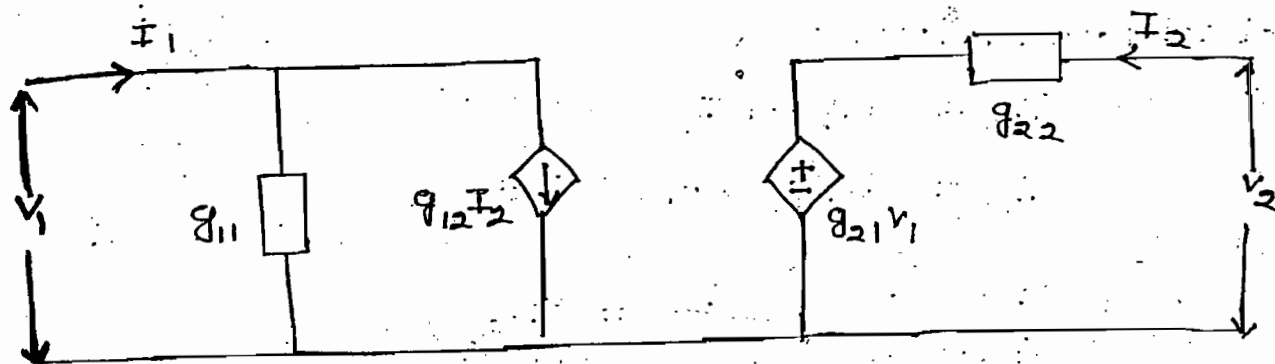
$$\left(g_{12} = \frac{Y_{12}}{Y_{22}} \right)$$

$$g_{22} = \left. \frac{V_2}{I_2} \right|_{V_1=0}$$

↓
Ω

$$\left(g_{22} \neq Z_{22}, \quad g_{22} = \frac{1}{Y_{22}} \right)$$

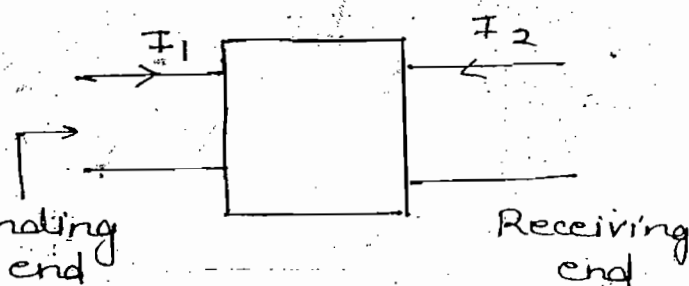
(mho)



ABCS Parameters:-

$$V_1 = AV_2 - BI_2 \quad \text{--- (I)}$$

$$I_1 = CV_2 - DI_2 \quad \text{--- (II)}$$



$$A = \left. \frac{V_1}{V_2} \right|_{I_2=0} \quad \left(A = \frac{Z_{11}}{Z_{21}} \right)$$

Unitless

$$C = \left. \frac{I_1}{V_2} \right|_{I_2=0} \quad \left(C = \frac{1}{Z_{21}} \right)$$

mho

$$B = \left. \frac{-V_1}{I_2} \right|_{V_2=0} \quad \left(B = -\frac{1}{Y_{21}} \right)$$

Ω

$$D = \left. \frac{-I_1}{I_2} \right|_{V_2=0} \quad \left(D = -\frac{Y_{11}}{Y_{21}} \right)$$

Unitless

By using eq. (I) & (II) it is not possible to develop ckt since both equations are developed w.r.t i/p

abcd Parameter:-

$$V_2 = aV_1 - bI_1 \quad \text{--- (I)}$$

$$I_2 = cV_1 - dI_1 \quad \text{--- (II)}$$

$$a = \left. \frac{V_2}{V_1} \right|_{I_1=0} \quad \left(a = \frac{Z_{22}}{Z_{12}} \right)$$

Unitless

$$c = \left. \frac{I_2}{V_1} \right|_{I_1=0} \quad \left(c = \frac{1}{Z_{12}} \right)$$

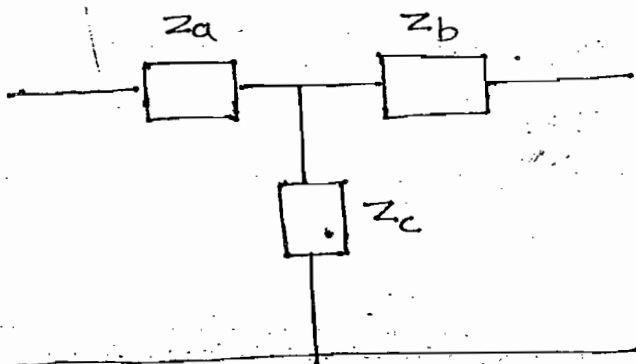
mho

$$b = \left. \frac{-V_2}{I_1} \right|_{V_1=0} \quad \left(b = -\frac{1}{Y_{12}} \right)$$

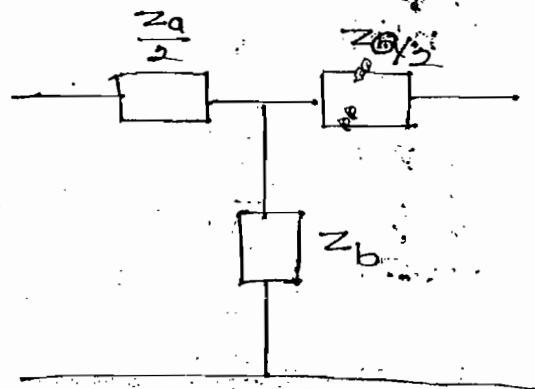
Ω

$$d = \left. \frac{-I_2}{I_1} \right|_{V_1=0} \quad \left(d = -\frac{Y_{22}}{Y_{12}} \right)$$

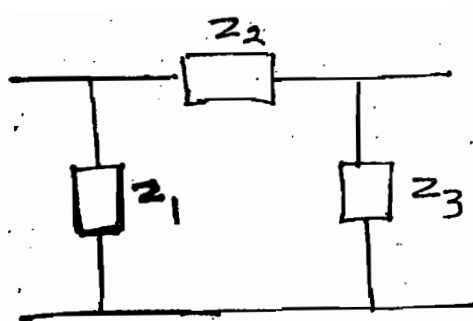
No units



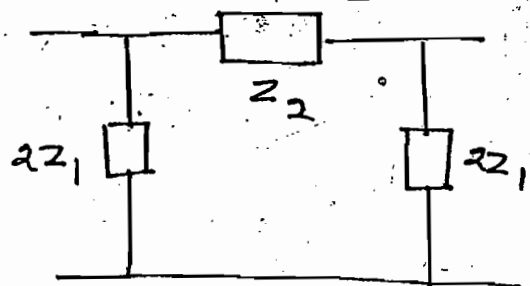
→ Unsymmetrical T-N/w



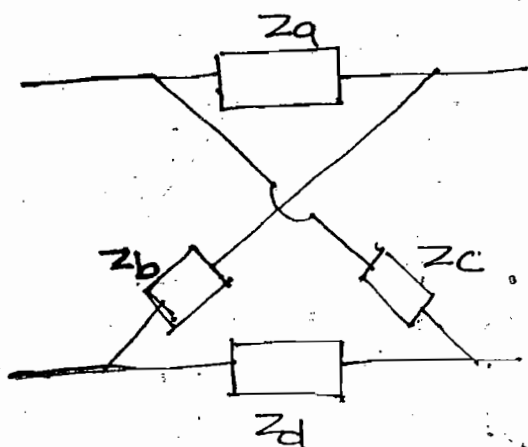
→ Symmetrical T-N/w



→ Unsymmetrical
T N/w



→ Symmetrical - T N/w



$Z_a = Z_d$
 $Z_b = Z_c$ } Symmetrical
Lattice
N/w

$Z_a \neq Z_d$
 $Z_b \neq Z_c$ } Unsymmetrical
Lattice N/w

**

Symmetrical	Reciprocal
$Z_{11} = Z_{22}$	$Z_{12} = Z_{21}$
$Y_{11} = Y_{22}$	$Y_{12} = Y_{21}$
$h_{11}h_{22} - h_{12}h_{21} = 1$	$h_{12} = -h_{21}$
$A = D$	$AD - BC = 1$

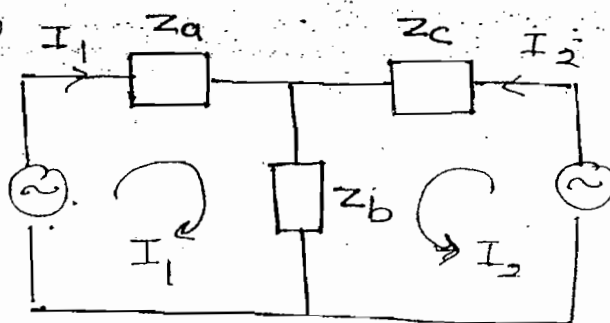
Z-Parameter → Use T-N/w

$$V_1 = (Z_a + Z_b)I_1 + Z_b I_2 \quad \text{--- (I)}$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (II)}$$

$$Z_{11} = Z_a + Z_b$$

$$Z_{12} = Z_b$$



$$V_2 = Z_b I_1 + (Z_b + Z_c) I_2 \quad \text{--- (III)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (IV)}$$

$$Z_{21} = Z_b \quad \& \quad Z_{22} = Z_b + Z_c$$

$$Z_{11} = Z_a + Z_b$$

$$Z_{11} = Z_a + Z_{12}$$

$$Z_a = Z_{11} - Z_{12}$$

$$Z_c = Z_{22} - Z_{12}$$

$$Z_b = Z_{12} = Z_{21}$$

For Y-Parameter $\rightarrow$ Convert complex N/w into Π -N/w

$$I_1 = V_1 Y_a + (V_1 - V_2) Y_b$$

$$I_1 = (Y_a + Y_b) V_1 - Y_b V_2 \quad \text{--- (I)}$$

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \text{--- (II)}$$

$$Y_{11} = Y_a + Y_b$$

$$Y_{12} = -Y_b$$

$$I_2 = V_2 Y_c + (V_2 - V_1) Y_b$$

$$\Rightarrow I_2 = -V_1 Y_b + (Y_b + Y_c) V_2 \quad \text{--- (III)}$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \text{--- (IV)}$$

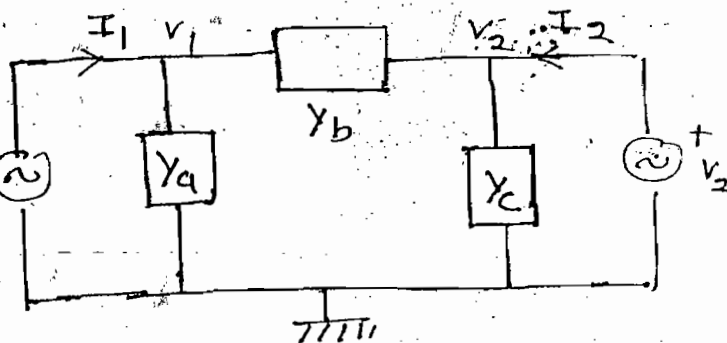
$$Y_{21} = -Y_b \quad \& \quad Y_{22} = Y_b + Y_c$$

**

$$Y_a = Y_{11} + Y_{12}$$

$$Y_c = Y_{22} + Y_{12}$$

$$Y_b = -Y_{12} = -Y_{21}$$



$$Z_a = Z_d, Z_b = Z_c$$

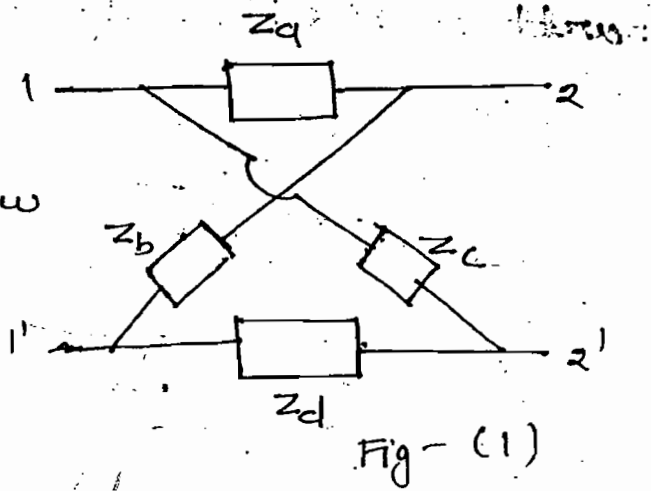
→ Symmetrical lattice N/w

$$Z_{11} = Z_{22} = \frac{Z_b + Z_a}{2}$$

$$Z_{12} = Z_{21} = \frac{Z_b - Z_a}{2}$$

$$Z_b = Z_{11} + Z_{12}$$

$$Z_a = Z_{11} - Z_{12}$$

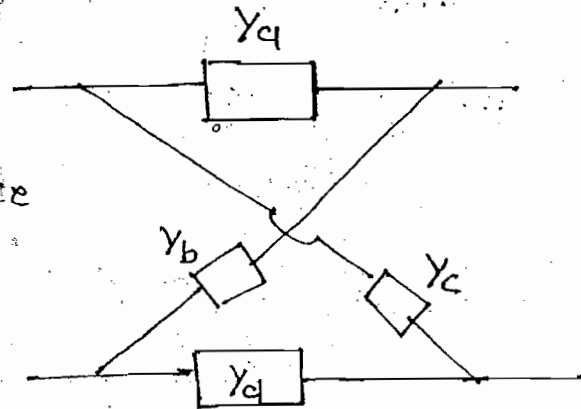


$$Y_a = Y_d, Y_b = Y_c$$

→ Symmetrical lattice N/w

$$Y_{11} = Y_{22} = \frac{Y_b + Y_a}{2}$$

$$Y_{12} = Y_{21} = \frac{Y_b - Y_a}{2}$$

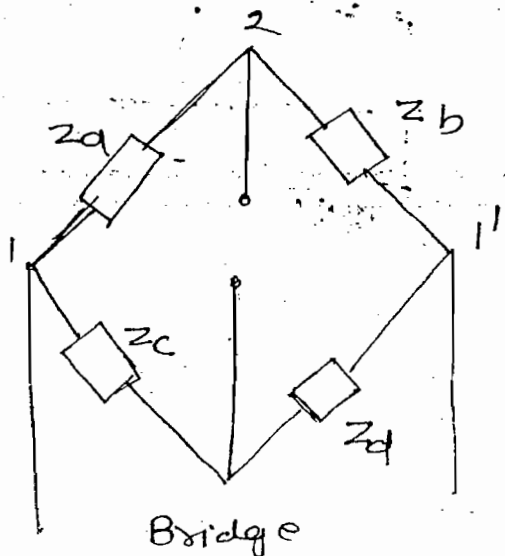


$$Y_b = Y_{11} + Y_{12}$$

$$Y_a = Y_{11} - Y_{12}$$

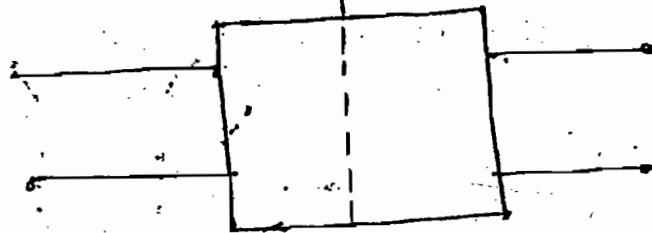
→ Fig-(1) & (ii) are same.

Fig-(ii)



Bartlett's Bi-section theorem:-

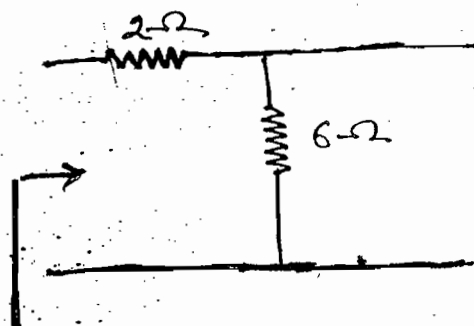
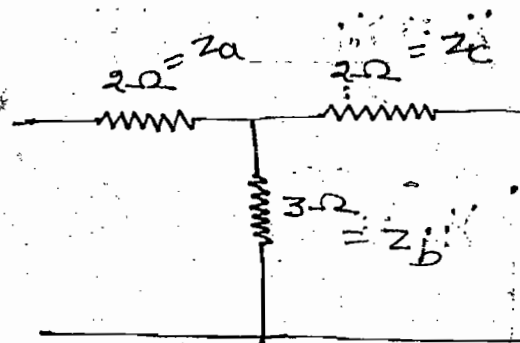
eg:- Filter



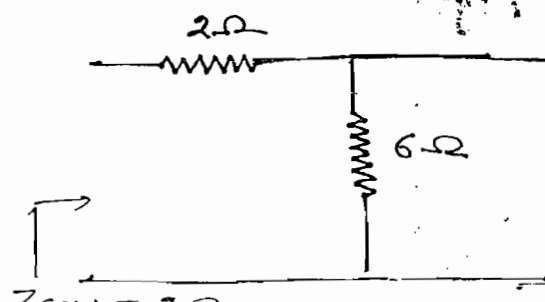
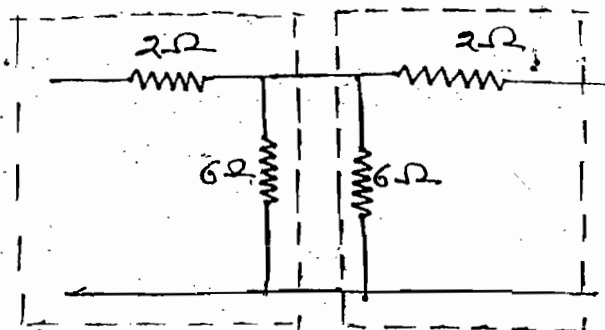
Symmetrical

$$Z_{11} = Z_{22} = \frac{Z_{OCH} + Z_{SCH}}{2}$$

$$Z_{12} = Z_{21} = \frac{Z_{OCH} - Z_{SCH}}{2}$$



$$Z_{OCH} = 2 + 6 = 8$$



$$Z_{11} = Z_{22} = \frac{Z_{OCH} + Z_{SCH}}{2} = \frac{8+2}{2} = 5$$

$$Z_{12} = Z_{21} = \frac{Z_{OCH} - Z_{SCH}}{2} = \frac{8-2}{2} = 3$$

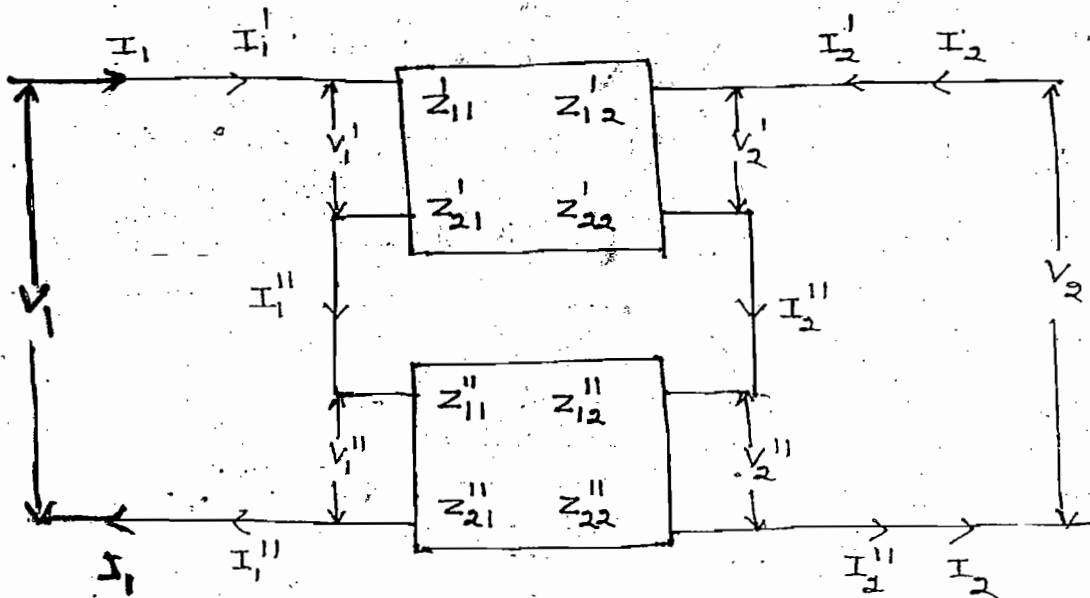
Alternate Way:-

$$Z_{11} = Z_a + Z_b = 5$$

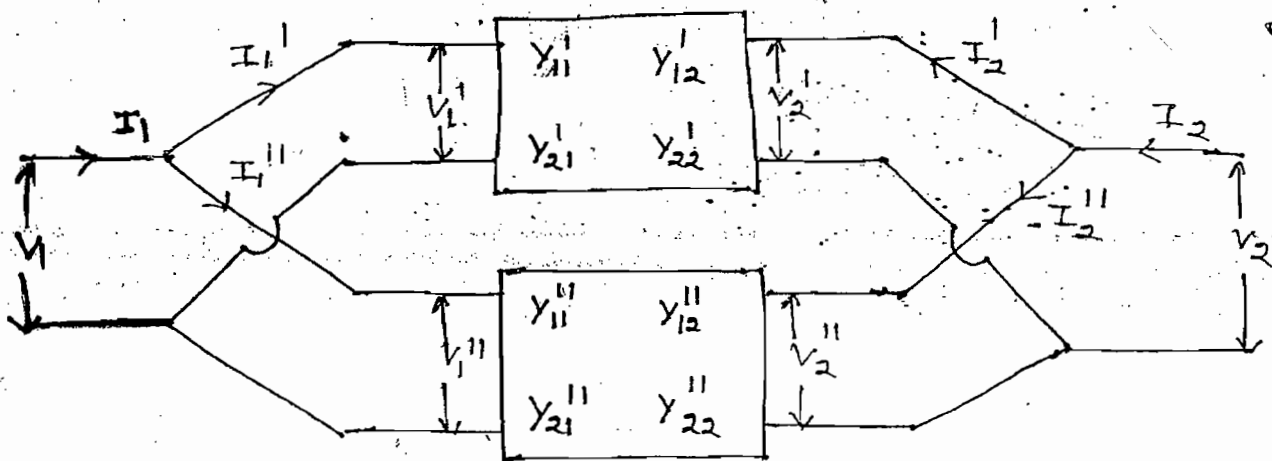
$$Z_{22} = Z_b + Z_c = 5$$

$$Z_{12} = Z_{21} = Z_b = 3$$

Series combination:-



Parallel combination:-



For series combination:-

$$V_1 = V_1' + V_1''$$

$$V_2 = V_2' + V_2''$$

$$I_1 = I_1' = I_1''$$

$$I_2 = I_2' = I_2''$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} Z_{11}' & Z_{12}' \\ Z_{21}' & Z_{22}' \end{bmatrix} + \begin{bmatrix} Z_{11}'' & Z_{12}'' \\ Z_{21}'' & Z_{22}'' \end{bmatrix}$$

For Parallel combination:-

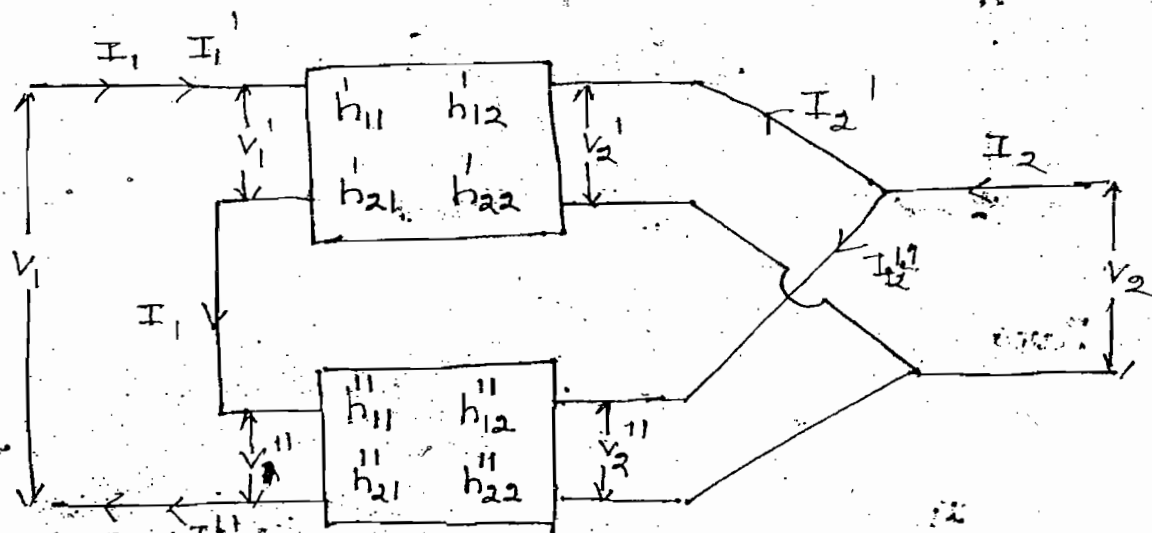
$$V_1 = V_1' = V_1''$$

$$V_2 = V_2' = V_2''$$

$$I_1 = I_1' + I_1''$$

$$I_2 = I_2' + I_2''$$

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} Y_{11}' & Y_{12}' \\ Y_{21}' & Y_{22}' \end{bmatrix} + \begin{bmatrix} Y_{11}'' & Y_{12}'' \\ Y_{21}'' & Y_{22}'' \end{bmatrix}$$



$$V_1 = V_1' = V_1''$$

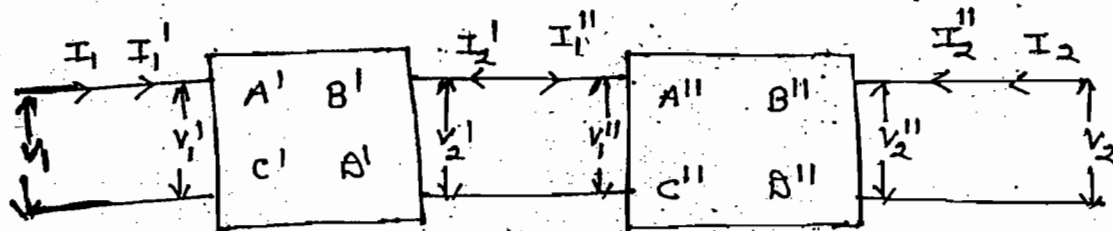
$$V_2 = V_2' = V_2''$$

$$I_1 = I_1' + I_1''$$

$$I_2 = I_2' + I_2''$$

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} h_{11}' & h_{12}' \\ h_{21}' & h_{22}' \end{bmatrix} + \begin{bmatrix} h_{11}'' & h_{12}'' \\ h_{21}'' & h_{22}'' \end{bmatrix}$$

Cascade or Tardum Connection:-



$$\begin{matrix} V_1 = V_1' \\ I_1 = I_1' \end{matrix}$$

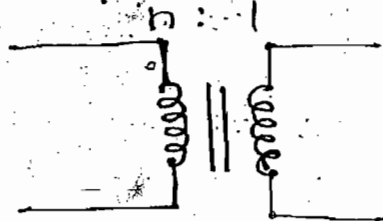
$$\begin{matrix} V_2' = V_1'' \\ I_1'' = -I_2' \end{matrix}$$

$$\begin{matrix} V_2 = V_2'' \\ I_2 = I_2'' \end{matrix}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix}$$

Ques:- Find $Z, Y, ABCD$ parameters of the N/w shown:-

Ideal t/f



Soln:-

$$A = \frac{V_1}{V_2} = n$$

$$B = -\frac{I_1}{I_2} = +\frac{1}{n}$$

-ve is due to current direction

$$\frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{n}{1}$$

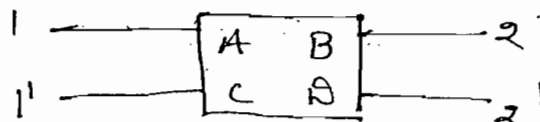
Note:-

In ideal t/f it is not possible to find impedance and admittance values since self and mutual inductance of ideal transformer are ∞

Ques:- Find O.C & S.C impedance w.r.t

(i) 1-1'

(ii) 2-2'



Soln:- (1) 1-1':-

$$Z_{i/p} = \frac{V_1}{I_1} = \frac{AV_2 - BI_2}{CV_2 - BI_2}$$

$$(Z_{i/p})_{o.c} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = \frac{A}{C} \Omega$$

$$(Z_{i/p})_{s.c} = \left. \frac{V_1}{I_1} \right|_{V_2=0} = \frac{B}{A} \Omega$$

(ii) 2-2':-

$$(Z_{o/p})_{o.c} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$

$$I_1 = CV_2 - BI_2 = 0$$

$$\Rightarrow \frac{V_2}{I_2} = \frac{B}{C}$$

$$(Z_{o/p})_{o.c} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = \frac{B}{C} \Omega$$

$$(Z_{o/p})_{s.c} = \left. \frac{V_2}{I_2} \right|_{V_1=0}$$

$$V_1 = AV_2 - BI_2$$

$$\Rightarrow 0 = AV_2 - BI_2$$

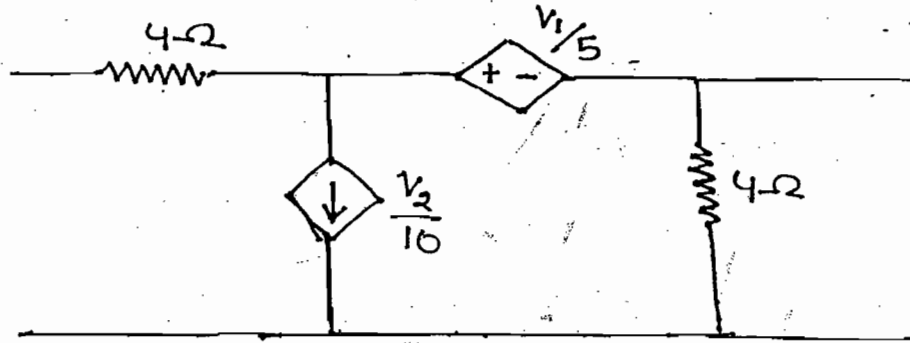
$$\Rightarrow \frac{V_2}{I_2} = \frac{B}{A}$$

$$(Z_{o/p})_{s.c} = \frac{B}{A} \Omega, \text{ Ans.}$$

$$B = - \frac{V_1}{I_2} \bigg|_{V_2=0} \Rightarrow \frac{2s+1}{s^2} = \frac{2V-(1)}{2V-(1)}$$

$$\Rightarrow \boxed{Z(s) = \frac{1}{s}} \quad \text{Ans}$$

ques:- Find B and A of the circuit shown

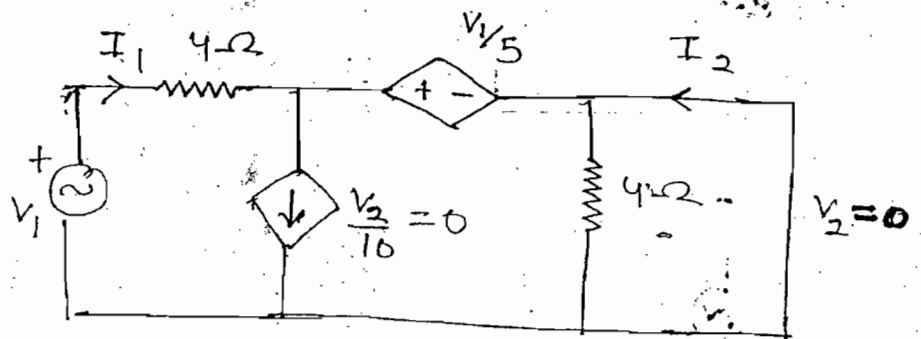


Soln:-

$$B = - \frac{V_1}{I_2} \bigg|_{V_2=0}$$

$$A = - \frac{I_1}{I_2} \bigg|_{V_2=0}$$

$$= - \frac{-(-I_2)}{I_2} = 1$$



$$I_1 = -I_2$$

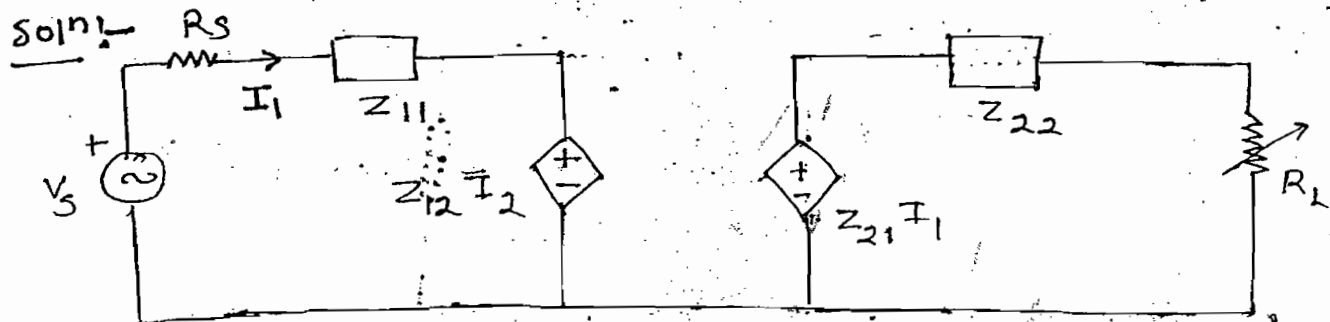
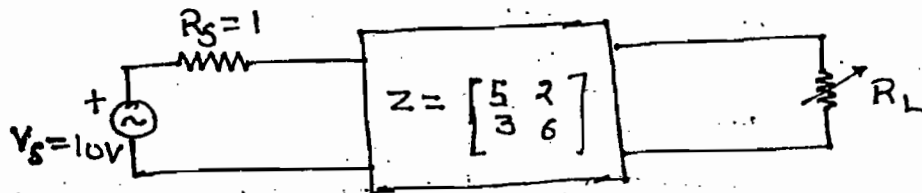
$$I_1 = \frac{V_1 - \frac{V_1}{5}}{4}$$

$$-I_2 = \frac{4V_1/5}{4}$$

$$\Rightarrow \boxed{B = - \frac{V_1}{I_2} = 5} \quad \text{Ans}$$

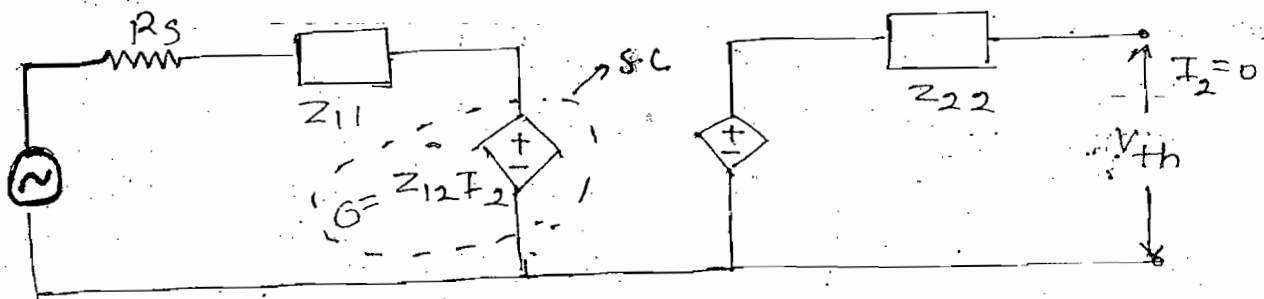
Lecture-12

ques:- Find max. power dissipation in load resistor



Case-(1):- V_{th}

Disconnect load resistor



$$I_1 = \frac{V_s}{R_s + Z_{11}} \quad \text{--- (I)}$$

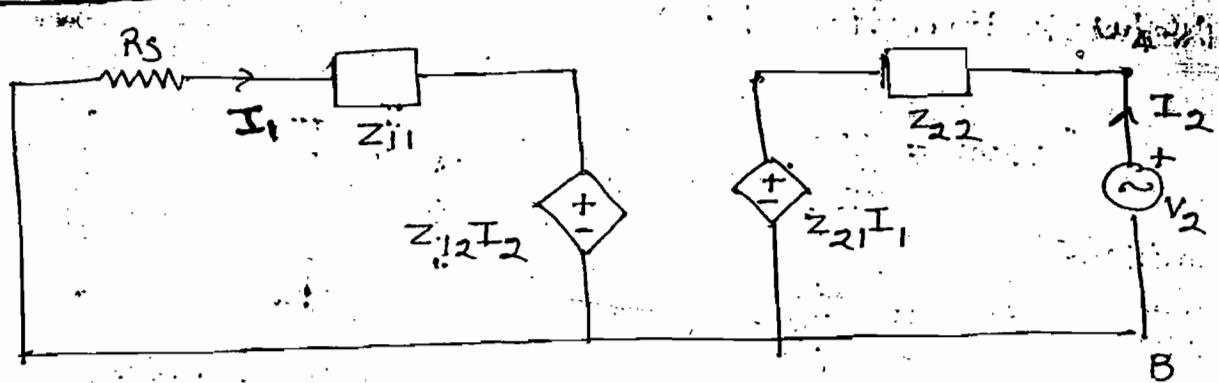
$$V_{th} = Z_{21} I_1 \quad \text{--- (II)}$$

Substitute eq-(I) in eq-(II)

$$V_{th} = Z_{21} \frac{V_s}{R_s + Z_{11}}$$

$$V_{th} = \frac{3 \times 10}{1 + 5} = 5$$

Case-(ii) (R_{th}): —



$$Z_{th} = \frac{V_2}{I_2}$$

$$I_2 = \frac{V_2 - z_{21}I_1}{z_{22}}$$

$$I_1 = -\frac{z_{12}I_2}{R_s + z_{11}} \quad \text{--- (i)}$$

$$I_2 z_{22} = V_2 - z_{21}I_1$$

$$V_2 = I_2 z_{22} + z_{21}I_1 \quad \text{--- (ii)}$$

Substitute eq-(i) in eq-(ii).

$$V_2 = I_2 z_{22} - \frac{z_{12} z_{21} I_2}{R_s + z_{11}}$$

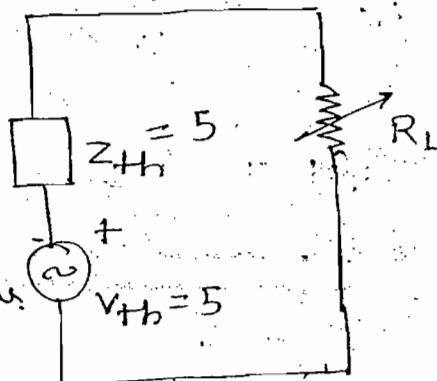
$$Z_{th} = \frac{V_2}{I_2} = z_{22} - \frac{z_{12} z_{21}}{R_s + z_{11}}$$

$$= 6 - \frac{2 \times 3}{1+5} \Rightarrow Z_{th} = 5$$

$$R_t = R_{th} = 5 \Omega$$

$$P_{max} = \frac{V_{th}^2}{4R_L}$$

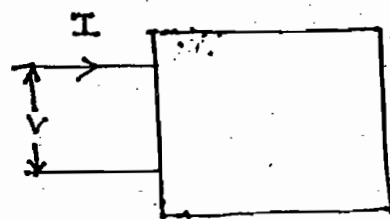
$$= \frac{25}{4 \times 5} = 1.25 \text{ W, Ans.}$$



Network Functions :-

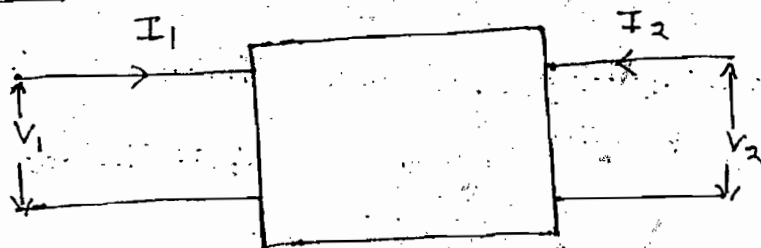
Immittance
Common Name

$$\begin{cases} Z(s) = \frac{V(s)}{I(s)} \\ Y(s) = \frac{I(s)}{V(s)} \end{cases}$$



Single Port

Two Port :-



Driving Point I/p impedance

$$(I) \quad Z_{11}(s) = \frac{V_1(s)}{I_1(s)} \quad Z_{22}(s) = \frac{V_2(s)}{I_2(s)}$$

$$(II) \quad Y_{11}(s) = \frac{I_1(s)}{V_1(s)} \quad Y_{22}(s) = \frac{I_2(s)}{V_2(s)} \rightarrow \text{Driving pt. admittance function}$$

$$(III) \quad Z_{12}(s) = \frac{V_1(s)}{I_2(s)} \quad Z_{21}(s) = \frac{V_2(s)}{I_1(s)} \rightarrow \text{Transfer impedance ratio}$$

$$(IV) \quad Y_{12}(s) = \frac{I_1(s)}{V_2(s)} \quad Y_{21}(s) = \frac{I_2(s)}{V_1(s)} \rightarrow \text{Transfer admittance ratio}$$

$$(V) \quad G_{12} = \frac{V_1(s)}{V_2(s)} \quad G_{21} = \frac{V_2(s)}{V_1(s)} \rightarrow \text{Transfer voltage ratio}$$

$$(VI) \quad \alpha_{12} = \frac{I_1(s)}{I_2(s)} \quad \alpha_{21} = \frac{I_2(s)}{I_1(s)} \rightarrow \text{Transfer current ratio}$$

→ N/w parameter is calculated at pre-defined condition (either o.c or s.c)

eg:- $Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0 \rightarrow \text{o.c}}$ $Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0 \rightarrow \text{s.c}}$

→ To find the N/w function, no pre-defined condition is required

→ To represent the N/w simultaneously four parameters are required

eg:- To develop T-N/w Z_{11}, Z_{12}, Z_{21} & Z_{22} are required.

→ By using only one N/w function it is possible to represent the N/w.

Network Synthesis:-

→ In the N/w analysis for a given N/w it is possible to find either voltage response or current response or impedance function or admittance function

→ In a N/w synthesis, for a given function N/w is designed (It is reverse procedure of N/w analysis)

→ In a N/w synthesis it is possible to design the N/w for the following functions

- (I) Single port $\rightarrow \begin{matrix} 1. Z(s) \\ 2. Y(s) \end{matrix} \Bigg\} \rightarrow \text{Driving Point Immittance function}$
- (II) Two port $\rightarrow \begin{matrix} 1. Z_{11}(s), Z_{22}(s), Y_{11}(s), Y_{22}(s) \\ 2. Z_{12}(s), Z_{21}(s), Y_{12}(s), Y_{21}(s), G_{12}, G_{21}, \alpha_{12}, \alpha_{21} \end{matrix} \Bigg\} \rightarrow \text{Transfer function}$

→ In the N/w synthesis for given function it is possible to design the following N/w

- (I) Series (II) Parallel (III) Ladder

(I) Series $\rightarrow$ Foster-I form

Partial fraction
of expansion

(II) Parallel $\rightarrow$ Foster-II form

(III) Ladder

Cauer-I form

Cauer-II form

Continued fraction
of expansion

Necessary Condition to design a N/w:-

(I) $F(s)$ should be ^{positive} ~~primary~~ real function (PRF)

$$\left. \begin{array}{l} R \geq 0 \\ L \geq 0 \\ C \geq 0 \end{array} \right\} \rightarrow \text{Positive Real}$$

(II) If $F(s)$ is PRF then $\frac{1}{F(s)}$ also PRF

(III) If $F_1(s)$ and $F_2(s)$ are PRF then

$F(s) = F_1(s) + F_2(s)$ are always PRF.

(IV) All the poles of the function should be present in the left half of the plane

(V) Imaginary poles and zeroes should be conjugate pairs

(VI) In the partial fraction of expansion residue should be positive real

(VII) Numerator and denominator polynomials should satisfy Hurwitz.

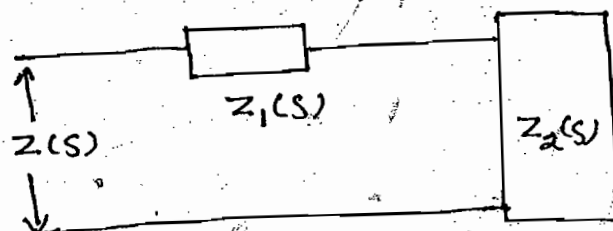
(VIII) The highest power of Numerator and denominator of polynomial should be differ at ^{at most} by unity. This condition prohibits multiple pole and zero at infinity.

9 → The lowest power of Numerator and Denominator should be differ^{by} at most unity. This condition prohibits multiple poles and zeroes at origin.

10 → Total No. of poles of a function should be equal to total no. of zeroes.

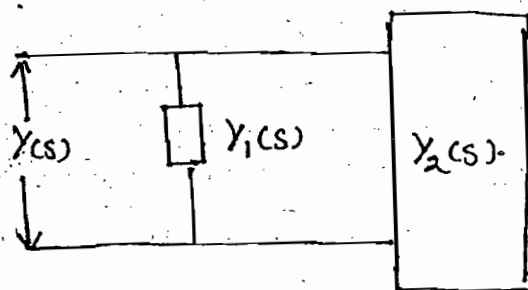
$$Z(s) = Z_1(s) + Z_2(s)$$

$$\Rightarrow Z_2(s) = Z(s) - Z_1(s)$$



$$Y(s) = Y_1(s) + Y_2(s)$$

$$\Rightarrow Y_2(s) = Y(s) - Y_1(s)$$



Removal of Pole at ∞ :-

$$Z(s) = \frac{b_{n+1}s^{n+1} + b_n s^n + \dots + b_0}{a_n s^n + a_{n-1}s^{n-1} + \dots + a_0}$$

Numerator Power > Denominator power
→ pole exist at ∞ .

By long division

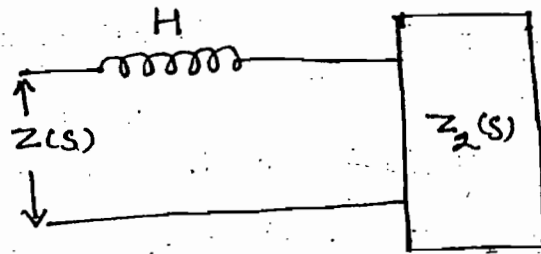
$$Z(s) = \frac{b_{n+1}s^{n+1}}{a_n s^n} + Z_2(s)$$

$$\left(H = \frac{b_{n+1}}{a_n} \right)$$

$$Z(s) = Hs + Z_2(s)$$

$$X_L = Ls$$

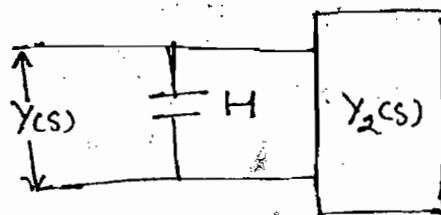
$$Z_2(s) = Z(s) - Hs$$



$$Y(s) = \frac{Hs}{s} + Y_2(s)$$

$$B_C = SC$$

$$Y_2(s) = Y(s) - HS$$



Removal of pole at origin:-

$$Z(s) = \frac{b_0 + \dots + b_{n-1}s^{n-1} + b_n s^n}{s(a_0 + \dots + a_{n-1}s^{n-1} + a_n s^n)}$$

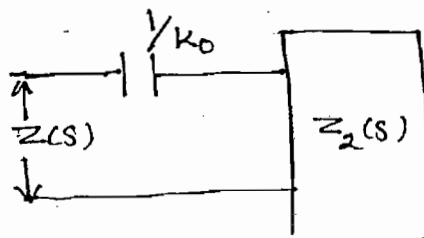
By long division

$$Z(s) = \frac{b_0}{s a_0} + Z_2(s) \quad \left(k_0 = \frac{b_0}{a_0} \right)$$

$$Z(s) = \frac{k_0}{s} + Z_2(s)$$

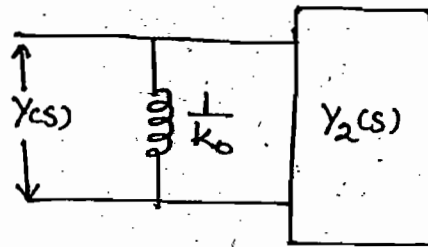
$$\Rightarrow Z_2(s) = Z(s) - \frac{k_0}{s}$$

$$X_C = \frac{1}{Cs}$$



$$Y(s) = \frac{k_0}{s} + Y_2(s)$$

$$Y_2(s) = Y(s) - \frac{k_0}{s}$$



Removal of conjugate Pair of Poles: —

$$Z(s) = Z_1(s) + Z_2(s)$$

$$Z_2(s) = Z(s) - Z_1(s)$$

$$Z_1(s) \rightarrow \text{Poles} \Rightarrow \pm j\omega$$

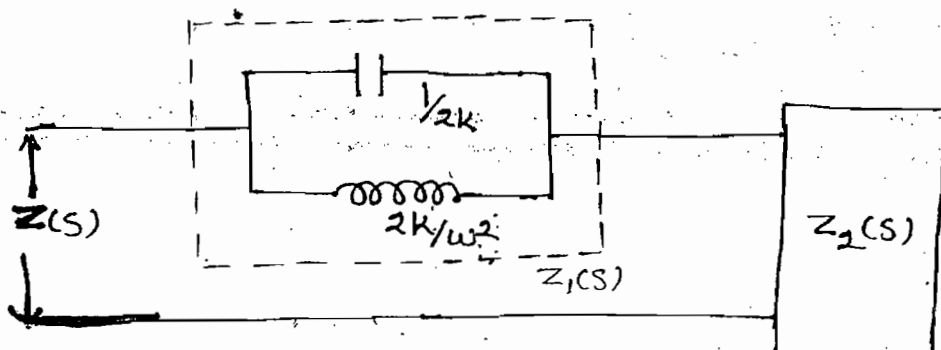
$$Z_1(s) = \frac{k_1}{s+j\omega} + \frac{k_2}{s-j\omega} \quad (k_1 = k_2 = k)$$

$$Z_1(s) = \frac{k}{s+j\omega} + \frac{k}{s-j\omega}$$

$$Z_1(s) = \frac{2ks}{s^2 + \omega^2}$$

$$Z_1(s) = \frac{1}{\frac{s^2}{2ks} + \frac{\omega^2}{2ks}} = \frac{1}{Y_a + Y_b}$$

$B_c = sC$ $B_h = \frac{1}{Ls}$

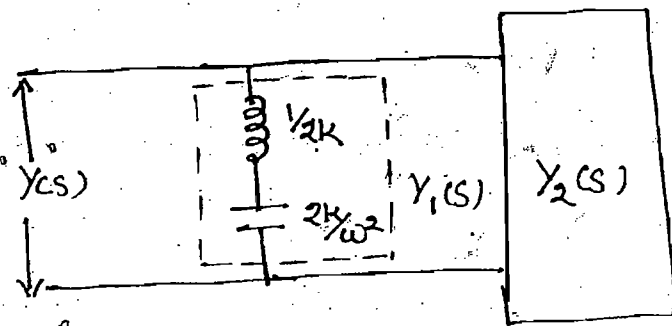


$$Y(s) = Y_1(s) + Y_2(s)$$

$$Y_2(s) = Y(s) - Y_1(s)$$

$$Y_1(s) = \frac{1}{\frac{s}{2k} + \frac{\omega^2}{2ks}} = \frac{1}{Z_a + Z_b}$$

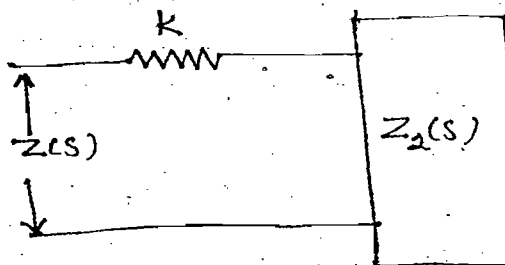
$X_L = sL$ $X_C = \frac{1}{Cs}$



Removal of constant!:-

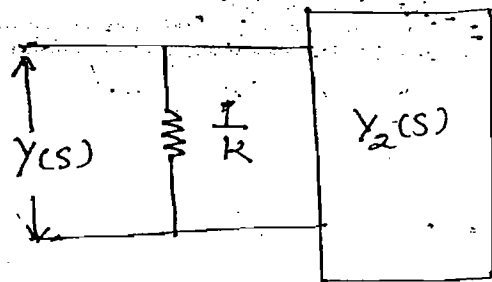
$$Z(s) = k + Z_2(s)$$

$$Z_2(s) = Z(s) - k$$



$$Y(s) = k + \frac{1}{Z_2(s)}$$

$$Y_2(s) = Y(s) - k$$

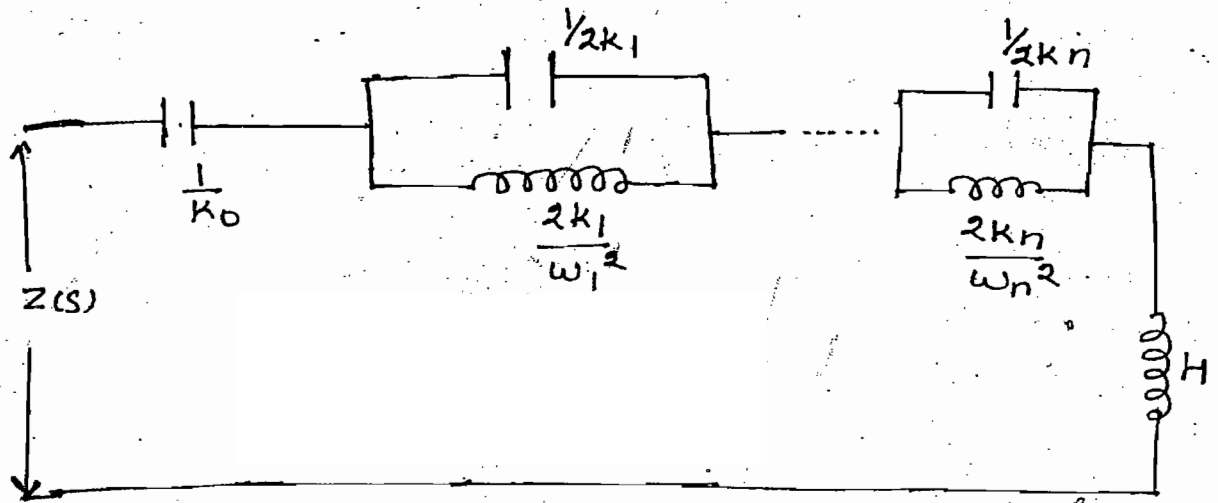


LC N/w Foster - I form :- (Series)

$$Z(s) = \frac{k_0}{s} + \sum_{i=1}^n \frac{2k_i s}{s^2 + \omega_i^2} + Hs$$

$\checkmark$
 $X_C = \frac{1}{Cs}$

$(X_L = Ls)$



Lecture-13

LC N/w Foster - II form :- (Parallel)

$$Y(s) = \frac{k_0}{s} + \sum_{i=1}^n \frac{2k_i s}{s^2 + \omega_i^2} + Hs$$

$\checkmark$
 $B_L = \frac{1}{Ls}$

$B_C = Sc$



Ques:- obtain foster - I & II from $Z(s)$

$$Z(s) = \frac{(s^2+2)(s^2+4)}{s(s^2+3)}$$

Soln:- Foster - I :-

Pole $\rightarrow \infty \rightarrow HS$
 Pole $\rightarrow \text{origin} \rightarrow k_0/s$
 Pole $\rightarrow \text{conjugate pair} \rightarrow \frac{2ks}{s^2 + \omega^2}$

$$Z(s) = \frac{k_0}{s} + \frac{2ks}{s^2 + \omega^2} + HS$$

$$\frac{(s^2+2)(s^2+4)}{s(s^2+3)} = \frac{k_0}{s} + \frac{2ks}{s^2+3} + HS \quad (\text{Partial Fraction})$$

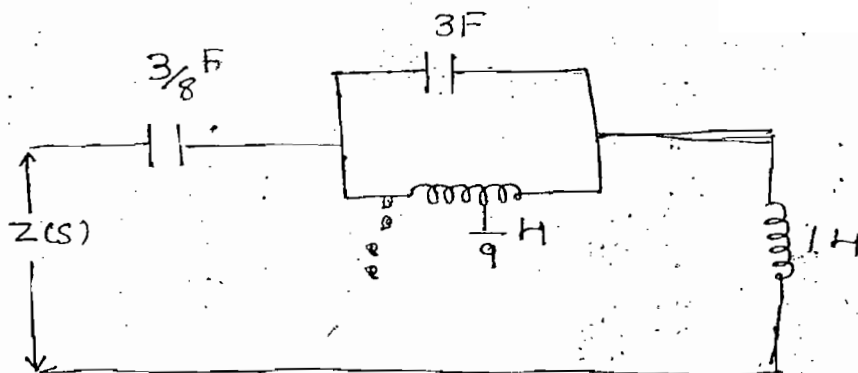
$$k_0 = \frac{8}{3} \quad 2k \rightarrow \frac{1}{3} \quad H \rightarrow 1$$

$$Z(s) = \frac{8/3}{s} + \frac{s/3}{s^2+3} + s$$

$$\Rightarrow Z(s) = \frac{1}{\frac{3}{8}s} + \frac{1}{\frac{s^2}{\frac{3}{8}} + \frac{3}{\frac{3}{8}}} + s$$

$$\Rightarrow Z(s) = \frac{1}{\frac{3}{8}s} + \frac{1}{\frac{3s+9}{s}} + s$$

$\downarrow \quad \downarrow \quad \downarrow$
 $X_C = \frac{1}{Cs} \quad B_C = sC \quad B_L = \frac{1}{sL}$
 $\quad \quad \quad \frac{1}{X_A + Y_B}$



Foster - II :-

Note :- If $F(s)$ is given by using same function -
 foster - I and II form are obtained.

$\rightarrow$ If $Z(s)$ is given by using $\frac{1}{Z(s)}$ function foster-II form is obtained

$\rightarrow$ If $Y(s)$ is given by using $\frac{1}{Y(s)}$ function foster-I form is obtained.

$$Y(s) = \frac{s(s^2+3)}{(s^2+2)(s^2+4)}$$

$P \rightarrow \infty$ X

$P \rightarrow \text{origin}$ X

$P \rightarrow \text{conjugate pair}$ ✓

$$Y(s) = \frac{2k_1 s}{s^2 + \omega_1^2} + \frac{2k_2 s}{s^2 + \omega_2^2}$$

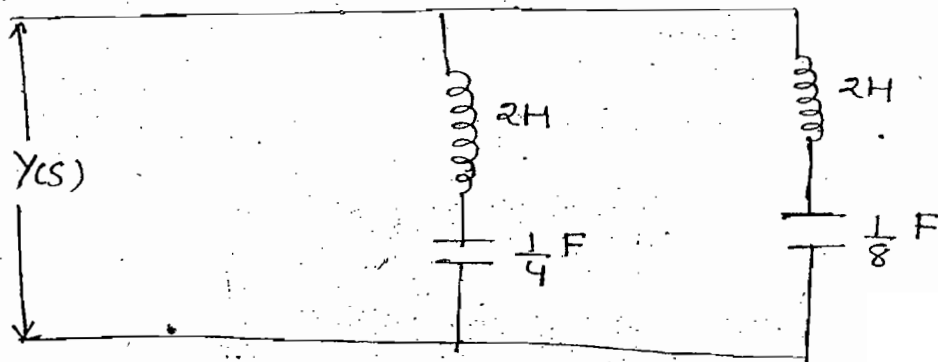
$$\frac{s(s^2+3)}{(s^2+2)(s^2+4)} = \frac{2k_1 s}{s^2+2} + \frac{2k_2 s}{s^2+4}$$

$$2k_1 \rightarrow \frac{1}{2} \quad \& \quad 2k_2 \rightarrow \frac{1}{2}$$

$$Y(s) = \frac{s/2}{s^2+2} + \frac{s/2}{s^2+4}$$

$$\Rightarrow Y(s) = \frac{1}{\frac{s^2}{s/2} + \frac{2}{s/2}} + \frac{1}{\frac{s^2}{s/2} + \frac{4}{s/2}}$$

$$\Rightarrow Y(s) = \frac{1}{\left(\begin{array}{c} 2s + \frac{4}{s} \\ x_L = Ls \quad x_C = \frac{1}{Cs} \end{array} \right) \frac{1}{Z_a + Z_b}} + \frac{1}{\left(\begin{array}{c} 2s + \frac{8}{s} \\ x_L \quad x_C \end{array} \right) \frac{1}{Z_c + Z_d}}$$



LC N/w Cauer-I form:-

→ Ladder

(1) Removal of Pole at ∞ :-

$$Z_2(s) = Z(s) - H_1 s \quad (X_L = Ls)$$

$$Y_2(s) = \frac{1}{Z_2(s)}$$

$$Y_3(s) = Y_2(s) - H_2 s \quad (B_C = Cs)$$

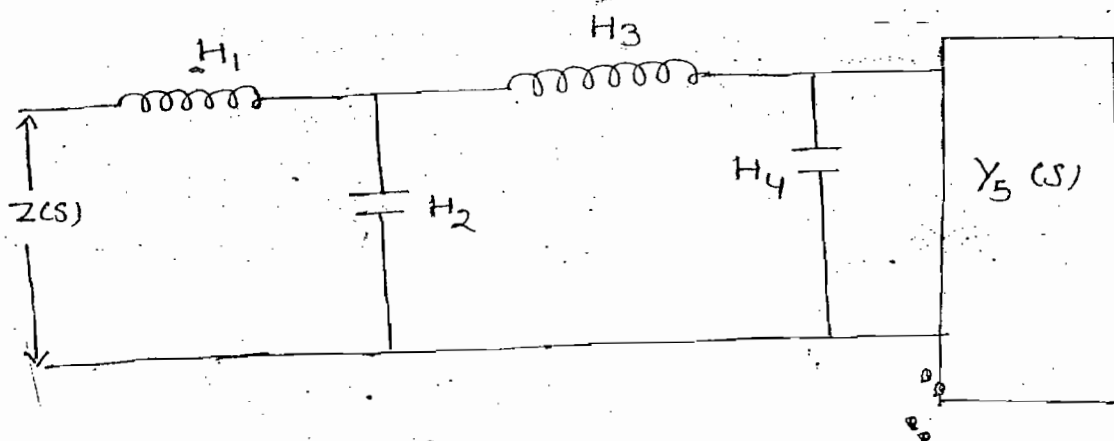
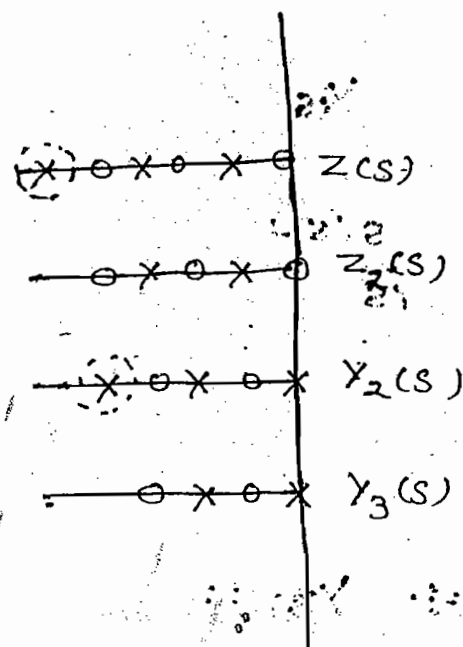
$$Z_3(s) = \frac{1}{Y_3(s)}$$

$$Z_4(s) = Z_3(s) - H_3 s$$

$$Y_4(s) = \frac{1}{Z_4(s)}$$

$$Y_5(s) = Y_4(s) - H_4 s$$

$$Y_4(s) = Y_5(s) + H_4 s$$



LC N/w Cauer-II form:-

Removal of Pole at origin:-

$$Z_2(s) = Z(s) - \frac{k_{01}}{s}$$

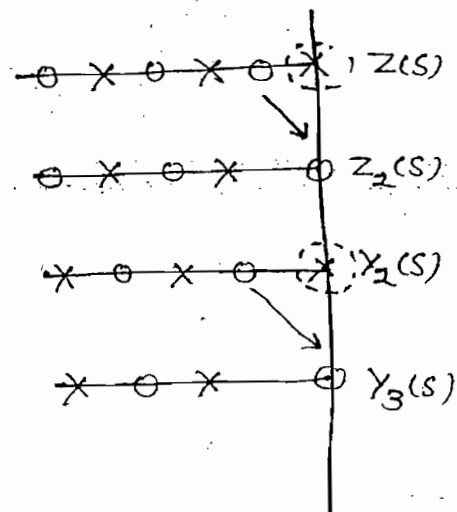
$$Y_2(s) = \frac{1}{Z_2(s)}$$

$$Y_3(s) = Y_2(s) - \frac{k_{02}}{s}$$

$$Z_3(s) = \frac{1}{Y_3(s)}$$

$$(X_C = \frac{1}{Cs})$$

$$(B_L = \frac{1}{Ls})$$

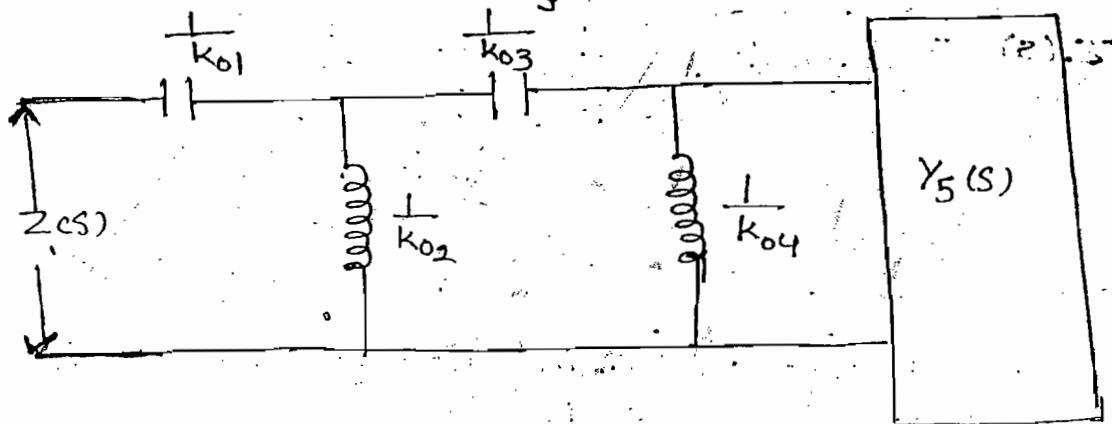


$$Z_4(s) = Z_3(s) - \frac{k_{03}}{s}$$

$$Y_4(s) = \frac{1}{Z_4(s)}$$

$$Y_5(s) = Y_4(s) - \frac{k_{04}}{s}$$

$$Y_4(s) = Y_5(s) + \frac{k_{04}}{s}$$



ques:- Obtain Cauer - I and II form

$$Z(s) = \frac{s^3 + 2s}{s^4 + 5s^2 + 4}$$

Soln:-

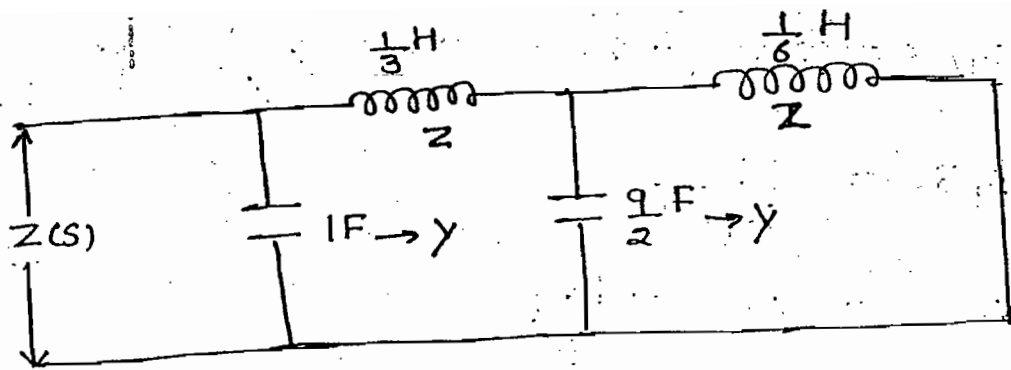
Cauer - I

Pole at $\infty \rightarrow$ not present

then reciprocal

$$Y(s) = \frac{s^4 + 5s^2 + 4}{s^3 + 2s}$$

$$\begin{aligned}
 & \frac{s^3 + 2s}{s^4 + 5s^2 + 4} \xrightarrow{\text{Long Division}} \frac{s^3}{s^4 + 5s^2 + 4} \rightarrow B_C = sC \rightarrow Y \\
 & \quad \frac{2s}{3s^2 + 4} \xrightarrow{\text{Long Division}} \frac{2s}{3s^2 + 4} \rightarrow X_L = Ls \rightarrow Z \\
 & \quad \frac{2s}{3s^2 + 4} \xrightarrow{\text{Long Division}} \frac{2s}{3s^2 + 4} \rightarrow B_C = sC \rightarrow Y \\
 & \quad \quad \frac{2s}{3} \xrightarrow{\text{Long Division}} \frac{2s}{3} \rightarrow X_L = Ls \rightarrow Z
 \end{aligned}$$



$$Z(s) = s + \frac{1}{\frac{s}{3} + \frac{1}{\frac{9s}{2} + \frac{1}{\frac{s}{6}}}}$$

Cauer - II Form:-

Pole $\rightarrow$ origin not present then take reciprocal

$$Z(s) = \frac{s(s^2+2)}{s^4+5s^2+4}$$

$$Y(s) = \frac{s^4+5s^2+4}{s^3+2s}$$

$$2s+s^3 \overline{) s^4+5s^2+4} \left(\frac{2}{s} \rightarrow B_L = \frac{1}{1s} \right)$$

$$\frac{2s^2+4}{3s^2+s^4} \overline{) 2s+s^3} \left(\frac{2}{3s} \rightarrow X_C = \frac{1}{6s} \right)$$

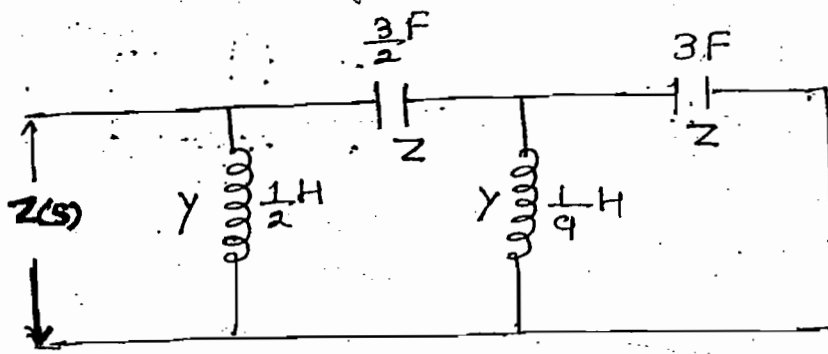
$$2s + \frac{2s^3}{3}$$

$$\frac{s^3}{3} \overline{) 3s^2+s^4} \left(\frac{9}{s} \rightarrow B_L \right)$$

$$\frac{s^3}{3} \overline{) \frac{s^3}{3}} \left(\frac{1}{3s} \rightarrow X_C \right)$$

$$\frac{s^3}{3}$$

X



$$Z(s) = \frac{3}{2s} + \frac{1}{\frac{2}{3s} + \frac{1}{\frac{9}{s} + \frac{1}{3s}}}$$

RC N/w Foster-I form:-

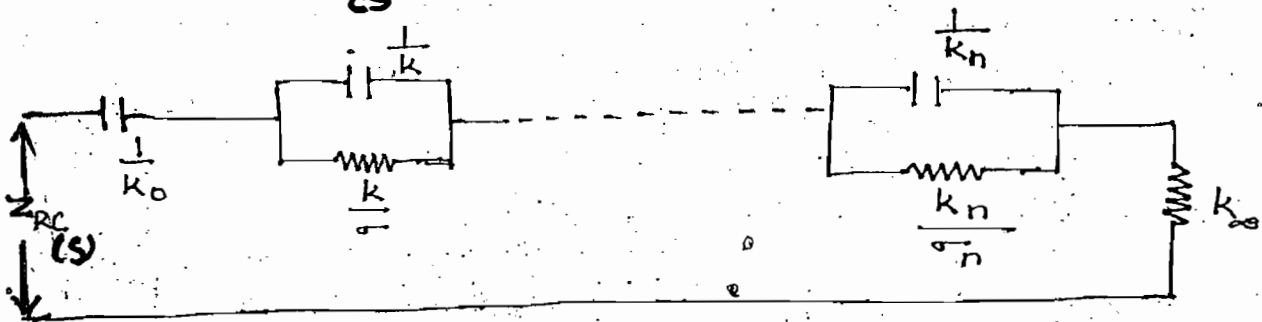
$$Z_{LC}(s) = \frac{k_0}{s} + \sum_{i=1}^n \frac{2k_i s}{s^2 + \omega_i^2} + Hs$$

$$Z_{RC}(s) = \frac{k_0}{s} + \sum_{i=1}^n \frac{k_i}{s + \sigma_i} + K_\infty$$

$$X_C = \frac{1}{Cs}$$

$$\frac{\frac{k}{s+\sigma}}{\frac{s}{k} + \frac{\sigma}{k}} = \frac{1}{X_a + Y_b}$$

$B_C = SC \quad G_1$

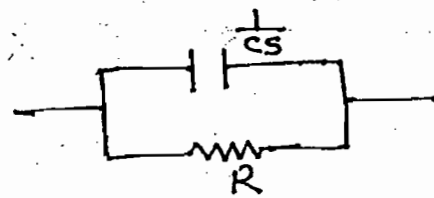


→ Series N/w is obtained

RC N/w Foster-II form:-

$$Z_{RC}(s) = \frac{k_0}{s} + \sum_{i=1}^n \frac{k_i}{s + \sigma_i}$$

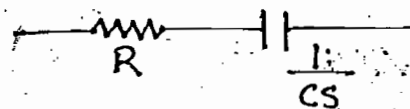
$$\rightarrow Z_{RC}(s) = \frac{R \frac{1}{Cs}}{R + \frac{1}{Cs}}$$



$$Z_{RC}(s) = \frac{R}{Rcs + 1}$$

$$Z_{RC}(s) = \frac{R}{RC(s + \frac{1}{RC})}$$

$$\rightarrow Y_{RC}(s) = \frac{1}{R + \frac{1}{Cs}}$$



$$Y_{RC}(s) = \frac{Cs}{Rcs + 1}$$

$$Y_{RC}(s) = \frac{Cs}{RC(s + \frac{1}{RC})}$$

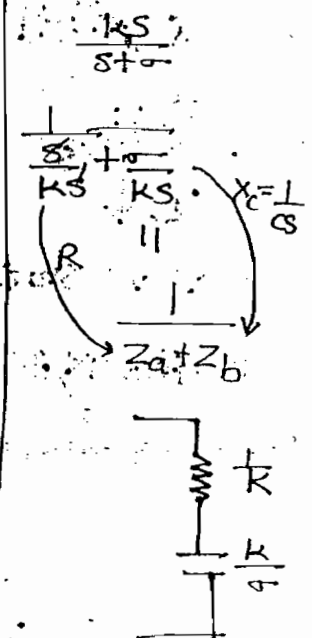
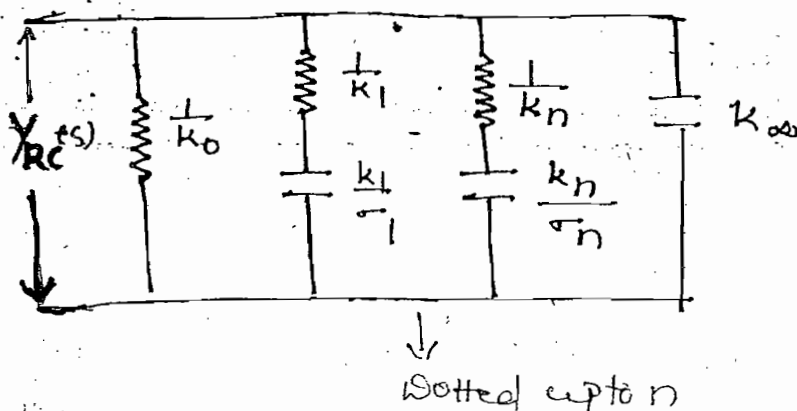
$$\frac{Y_{RC}(s)}{s} = \frac{1}{R(s + \frac{1}{RC})}$$

Note:-

→ Pole-zero pattern of Z_{RC} and Y_{RC} are identical. Thereby both functions can be represented by same mathematical equation.

$$\frac{Y_{RC}(s)}{s} = \frac{k_0}{s} + \sum_{i=1}^n \frac{k_i}{s + \sigma_i} + k_\infty$$

$$Y_{RC}(s) = k_0 + \sum_{i=1}^n \frac{k_i s}{s + \sigma_i} + k_\infty s$$



RC N/w Cauer - I form:-

$$LC \rightarrow Z_2(s) = Z(s) - H_1 s \quad Y_2(s) = \frac{1}{Z_2(s)}$$

($X_L = 1s$)

$$Y_3(s) = H_2 s$$

RC $\rightarrow$ Step-I:-

Removal of constant from $Z(s)$

$$Z_2(s) = Z(s) - K_1$$

$$Y_2(s) = \frac{1}{Z_2(s)}$$

Step-II:-

Removal of Pole at ∞ from $Y(s)$

$$Y_3(s) = Y_2(s) - H_1 s$$

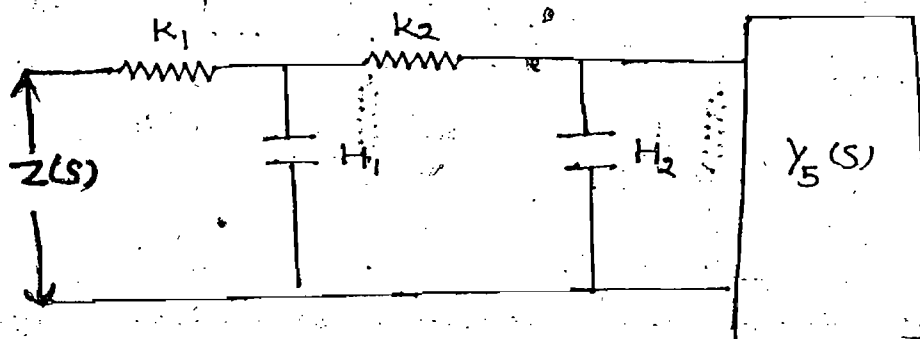
$$Z_3(s) = \frac{1}{Y_3(s)}$$

$$Z_4(s) = Z_3(s) - K_2$$

$$Y_4(s) = \frac{1}{Z_4(s)}$$

$$Y_5(s) = Y_4(s) - H_2 s$$

$$Y_4(s) = Y_5(s) + H_2 s$$



$\rightarrow$ ladder N/w is obtained

Note:-

Step-I and Step-II are alternatively repeated until the total function is realised

RC N/w Cauer-II form:-

$$LC \rightarrow Z_2(s) = Z(s) - \frac{k_{01}}{s} \quad Y_2(s) = \frac{1}{Z_2(s)} \quad Y_3(s) = \frac{k_{02}}{s}$$

$$(X_c = \frac{1}{Cs}) \quad (B_L = \frac{1}{Ls})$$

RC $\rightarrow$ Step-I:-

Removal of pole at origin from $Z(s)$

$$Z_2(s) = Z(s) - \frac{k_{01}}{s}$$

$$Y_2(s) = \frac{1}{Z_2(s)}$$

Step-II:-

Removal of constant from $Y(s)$

$$Y_3(s) = Y_2(s) - k_1$$

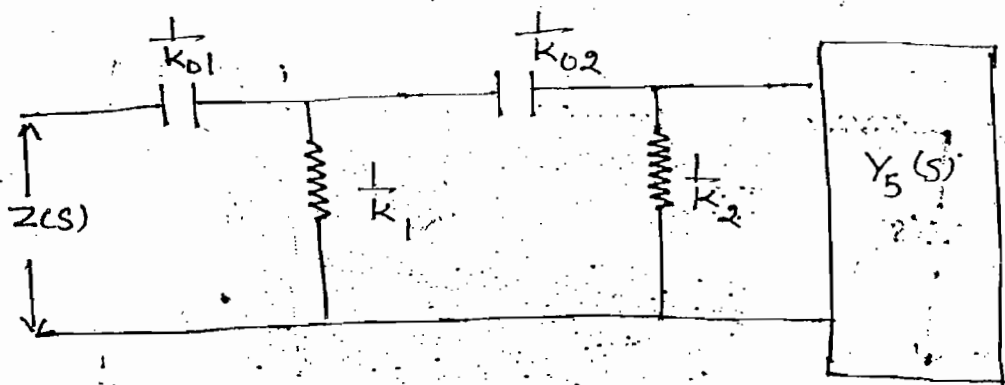
$$Z_3(s) = \frac{1}{Y_3(s)}$$

$$Z_4(s) = Z_3(s) - \frac{k_{02}}{s}$$

$$Y_4(s) = \frac{1}{Z_4(s)}$$

$$Y_5(s) = Y_4(s) - k_2$$

$$Y_4(s)' = Y_5(s) + k_2$$



$\rightarrow$ Ladder N/w is obtained.

Note:-

Step-I and II are alternatively repeated until the total function is realised.